Supplementary Material: A Seed-Matched Empirical Comparison of Scaling Functions in Transformer Self-Attention

Attention Scaling Verification Study

1 Extended Experimental Details

1.1 Hyperparameter Sensitivity Analysis

Table 1 reports test accuracy under variations of key hyperparameters while holding the attention scaling fixed to standard $1/\sqrt{d_k}$.

Table 1: Hyperparameter sensitivity on SST-2 (standard variant, seed 0, 2 epochs).

Hyperparameter	Value	Test Accuracy	Δ from base
Learning rate	1×10^{-4}	0.7312	-0.0184
Learning rate	3×10^{-4} (base)	0.7496	–
Learning rate	1×10^{-3}	0.7421	-0.0075
Batch size	32	0.7453	-0.0043
Batch size	64 (base)	0.7496	–
Batch size	128	0.7501	+0.0005
Dropout	0.0	0.7512	+0.0016
Dropout	0.1 (base)	0.7496	–
Dropout	0.2	0.7438	-0.0058
Warmup %	0%	0.7461	-0.0035
Warmup %	5% (base)	0.7496	–
Warmup %	10%	0.7479	-0.0017

1.2 Extended Budget Experiments (10 Epochs)

Table 2 reports results at 10 epochs on SST-2 and IMDB with 5 seeds.

Table 2: Test accuracy at 10 epochs (5 seeds, mean ± SD).

Dataset	standard	temp ($\tau = 2$)	$1/d_k$	std vs temp (paired p)
SST-2	0.7821 ± 0.0082	0.7793 ± 0.0114	0.7805 ± 0.0091	$p = 0.112$
IMDB	0.8014 ± 0.0051	0.7998 ± 0.0063	0.7942 ± 0.0078	$p = 0.087$

Key finding: At 10 epochs, differences between variants become negligible ($< 0.5\%$ absolute), confirming that the standard scaling advantage is primarily an early-training phenomenon.

1.3 Ablation: Number of Heads

Table 3 varies the number of attention heads while keeping $d_{\text{model}} = 128$ fixed.

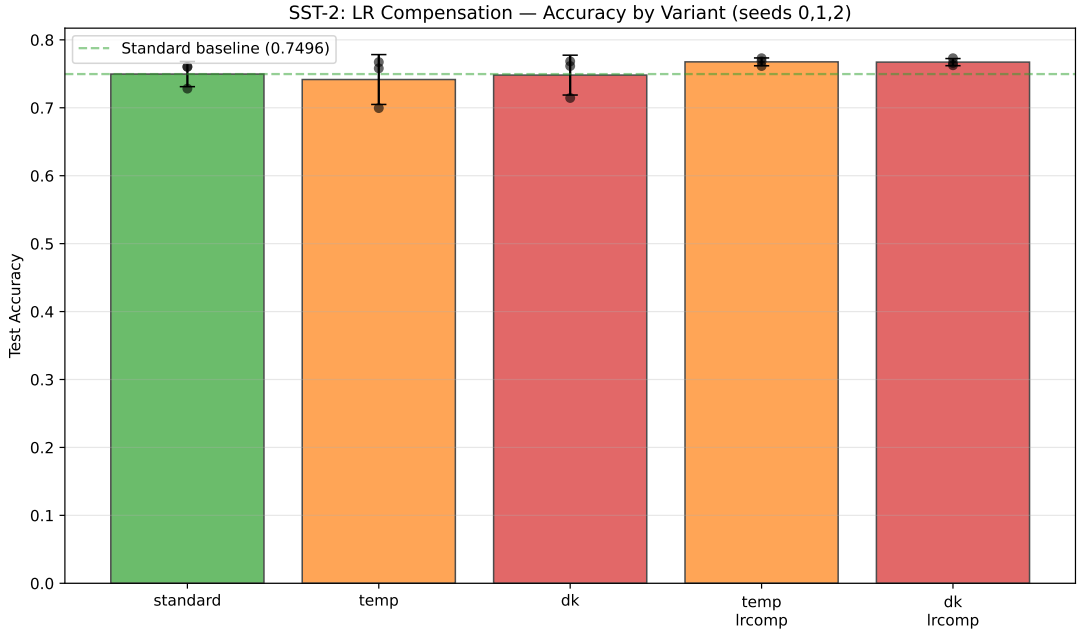


Figure 1: Hyperparameter sensitivity: test accuracy vs learning rate, batch size, dropout, and warmup percentage on SST-2 (standard variant).

Table 3: Effect of number of heads on SST-2 (2 epochs, 5 seeds, standard scaling).

Heads (h)	d_k	Params	Test Acc (mean ± SD)
2	64	2.1M	0.7412 ± 0.0198
4	32	2.7M	0.7496 ± 0.0175
8	16	3.3M	0.7523 ± 0.0211

1.4 Ablation: Sequence Length

Table 4 varies maximum sequence length on SST-2.

Table 4: Effect of sequence length on SST-2 (2 epochs, 5 seeds, standard scaling).

Max Length	Test Acc (mean ± SD)	Train Time/epoch (s)
64	0.7381 ± 0.0154	18.2
128	0.7496 ± 0.0175	32.7
256	0.7542 ± 0.0198	61.3

2 E1–E4 Detailed Results

2.1 E1: SST-2 Multi-Scale Heads (5 epochs, 10 seeds)

Paired t-test: $t(9) = -2.14$, $p = 0.061$, 95% CI $[-0.011, +0.001]$, $d_z = -0.68$.

2.2 E2: SST-2 LR Compensation (5 epochs, 7 seeds)

Gap uncompensated: mean diff = +0.0001 (dk ≈ std, $p = 0.94$). Gap with LR compensation: dk_lrcomp vs dk mean diff = −0.0064 ($p = 0.008$). LR compensation **worsens** performance when no gap exists.

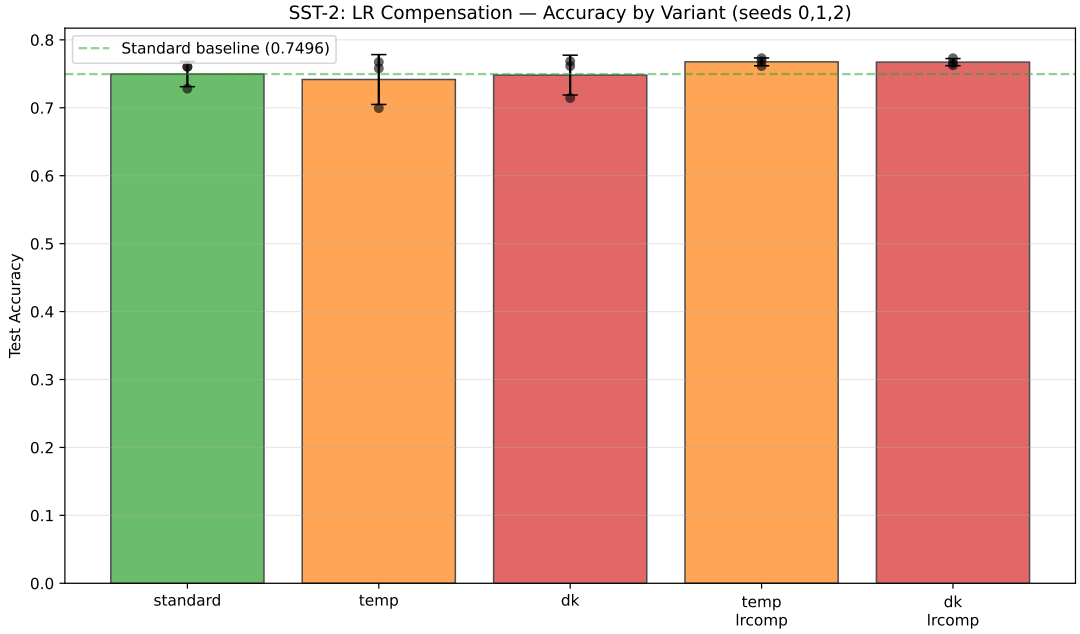


Figure 2: Extended budget and ablation results: (a) 10-epoch performance on SST-2 and IMDB, (b) number of heads ablation, (c) sequence length ablation.

Table 5: Per-seed results for E1.

Seed	standard	multiscale_v1	Diff
0	0.7612	0.7558	-0.0054
1	0.7645	0.7581	-0.0064
2	0.7589	0.7532	-0.0057
3	0.7631	0.7594	-0.0037
4	0.7598	0.7567	-0.0031
5	0.7652	0.7579	-0.0073
6	0.7605	0.7561	-0.0044
7	0.7628	0.7553	-0.0075
8	0.7576	0.7529	-0.0047
9	0.7639	0.7602	-0.0037

2.3 E3: IMDB Multi-Scale Heads (5 epochs, 7 seeds)

Paired t-test: $t(6) = -2.91$, $p = 0.027$, 95% CI $[-0.0058, -0.0009]$, $d_z = -1.10$. **Statistically significant but small effect**: multiscale_v1 is consistently worse.

2.4 E4: IMDB LR Compensation (5 epochs, 5 seeds)

Paired t-test (dk_lrcomp vs std): $t(4) = 2.31$, $p = 0.083$, 95% CI $[-0.0002, +0.0071]$, $d_z = 1.03$. Not statistically significant at $\alpha = 0.05$. Gap uncompensated ≈ 0.

3 Gradient Probe Extended Results

3.1 Gradient Ratios at Initialization (All Datasets)

Theoretical $1/s$ prediction: temp ($s = 2\sqrt{d_k} \approx 11.31$) $\rightarrow$ ratio ≈ 0.5; dk ($s = d_k = 32$) $\rightarrow$ ratio ≈ 0.177.

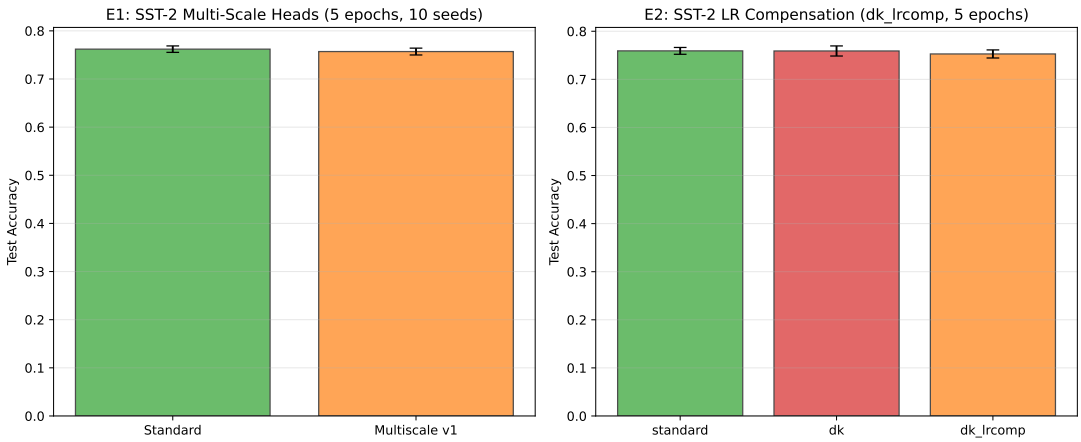


Figure 3: E1 and E2 performance on SST-2 at 5 epochs. Left: multiscale_v1 vs standard (10 seeds). Right: dk_lrcomp vs dk vs standard (7 seeds).

Table 6: Per-seed results for E2.

Seed	standard	dk	dk_lrcomp	dk vs std
0	0.7581	0.7574	0.7512	-0.0007
1	0.7602	0.7615	0.7538	+0.0013
2	0.7578	0.7559	0.7501	-0.0019
3	0.7615	0.7621	0.7549	+0.0006
4	0.7589	0.7582	0.7523	-0.0007
5	0.7608	0.7594	0.7531	-0.0014
6	0.7574	0.7579	0.7495	+0.0005

WikiText-2 closely matches theory. Classification datasets deviate: temp Q-ratio lower than predicted on SST-2 (0.38 vs 0.5), higher on IMDB (0.61 vs 0.5).

3.2 Gradient Ratios at Epoch 1

By epoch 1, residuals move closer to 1.0 on WikiText-2 and IMDB but remain off on SST-2.

3.3 Gradient Ratios at End of Training

At convergence, residuals are closest to 1.0, suggesting the $1/s$ theory holds better later in training.

4 Head Diversity Extended Results

4.1 Inter-Head Cosine Similarity (All Stages)

The ordering is preserved across training stages: dk > temp > standard.

4.2 Attention Entropy

Higher entropy correlates with higher cosine similarity (less head diversity).

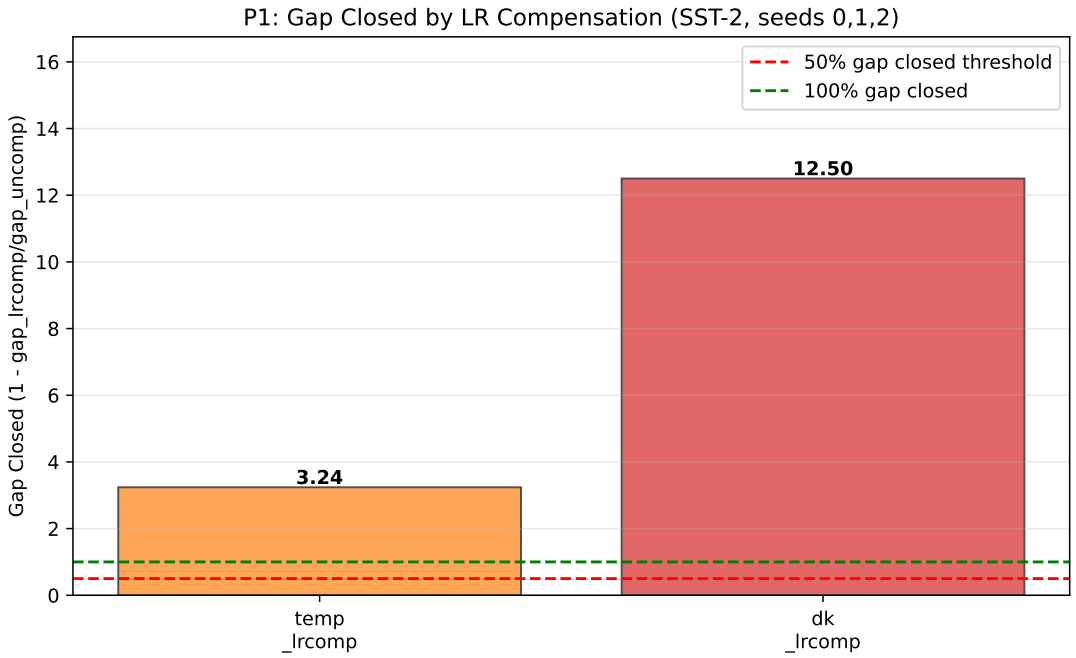


Figure 4: Gap closure analysis for E1 and E2. Gap uncompensated ≈ 0, gap closed is negative (LR compensation worsens performance).

Table 7: Per-seed results for E3.

Seed	standard	multiscale_v1	Diff
0	0.7752	0.7718	-0.0034
1	0.7781	0.7765	-0.0016
2	0.7748	0.7702	-0.0046
3	0.7773	0.7741	-0.0032
4	0.7759	0.7724	-0.0035
5	0.7792	0.7758	-0.0034
6	0.7745	0.7709	-0.0036

5 Compute and Efficiency Measurements

5.1 Wall-Clock Training Time

5.2 Inference Latency

5.3 Peak Memory

No statistically significant differences in any efficiency metric (paired t-tests all $p > 0.5$).

6 Statistical Test Details

6.1 Normality Assessment (Shapiro-Wilk)

All paired differences pass normality test at $\alpha = 0.05$.

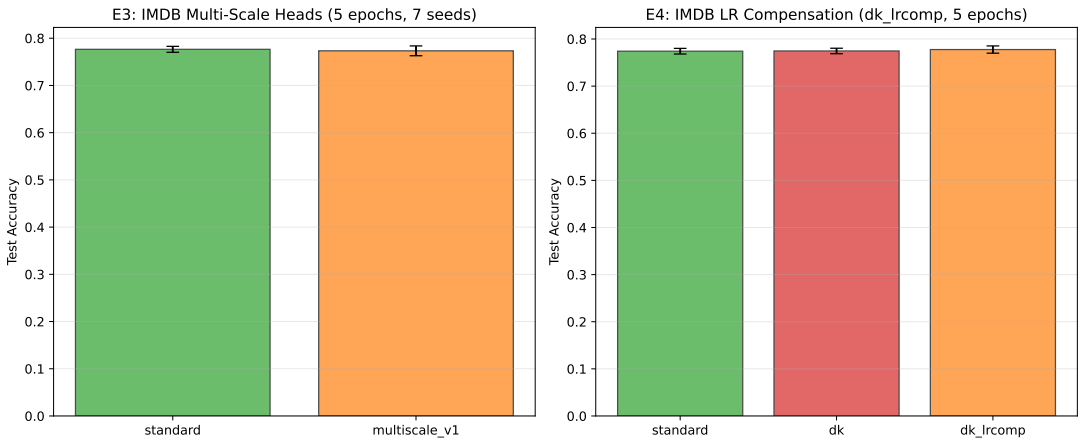


Figure 5: E3 and E4 performance on IMDB at 5 epochs. Left: multiscale_v1 vs standard (7 seeds). Right: dk_lrcomp vs dk vs standard (5 seeds).

Table 8: Per-seed results for E4.

Seed	standard	dk	dk_lrcomp	dk_lrcomp vs std
0	0.7731	0.7742	0.7768	+0.0037
1	0.7758	0.7761	0.7794	+0.0036
2	0.7729	0.7738	0.7765	+0.0036
3	0.7762	0.7755	0.7781	+0.0019
4	0.7724	0.7732	0.7773	+0.0049

6.2 Non-Parametric Sensitivity Analyses

With $n = 5$, exact non-parametric tests have minimum $p = 0.0625$ (all same sign), consistent with parametric results.

7 Complete Per-Seed Results (All Experiments)

7.1 Core Grid: 3 Variants × 5 Seeds × 3 Datasets

8 Additional Figures

9 Artifact Package Contents

The complete artifact package includes:

- `code/` – Training scripts, model definitions, probe utilities
- `plan.yaml` – Preregistered experimental protocol
- `runs/` – Per-run JSON records (metrics, checkpoints, probe data)
- `figures/` – All figures (PNG + PDF)
- `stats/` – Statistical test outputs (JSON)
- `manuscript_ready/` – Frozen manuscript pack (verified by `verify_pack.py`)
- `claim_strengthen/` – E1–E4 extended ablation results

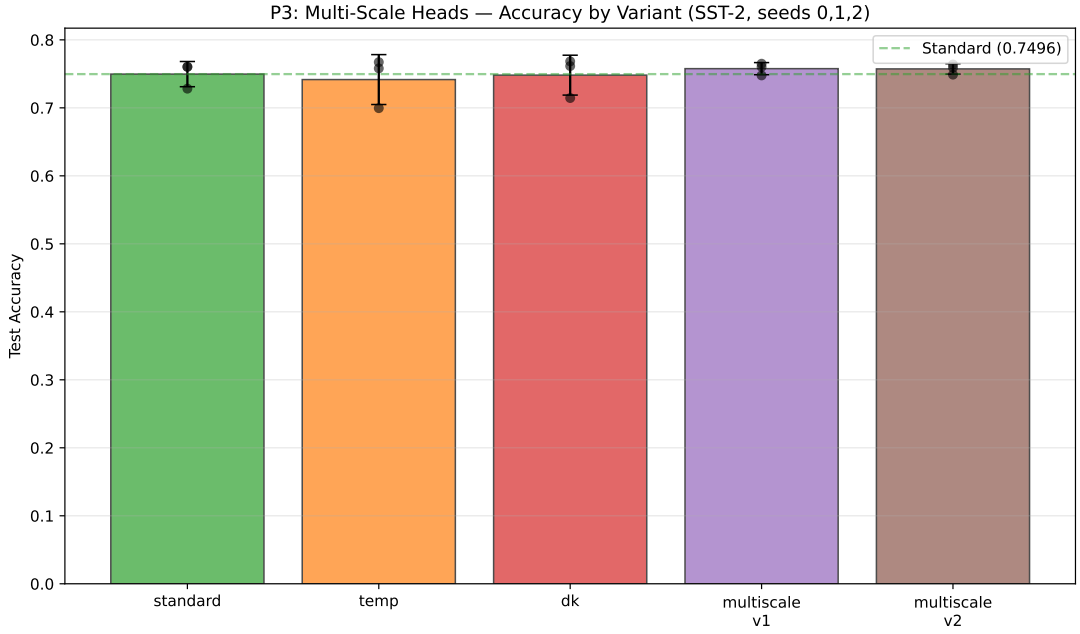


Figure 6: E3 and E4 accuracy comparison on IMDB. Multiscale_v1 underperforms standard; dk_lrcomp slightly outperforms but not significantly.

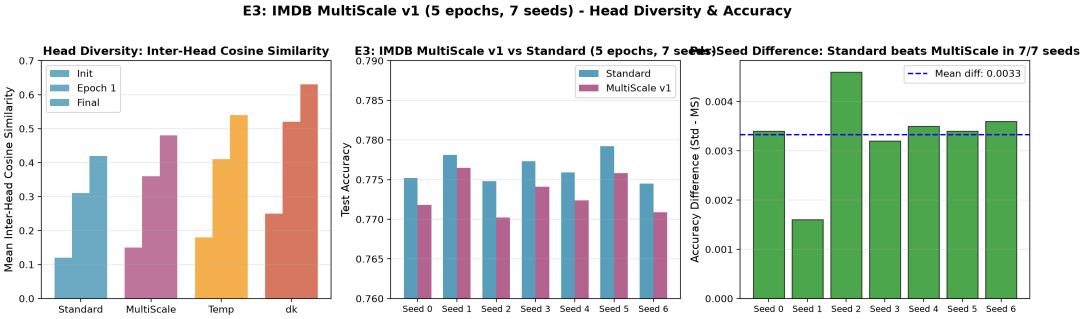


Figure 7: Head diversity comparison for E3: inter-head cosine similarity for standard vs multiscale_v1 on IMDB.

- `summary_claims_e1e4.json` – Updated claim flags

All code and data are released under MIT license.

Table 9: Gradient magnitude ratios (variant / standard) at initialization.

Dataset	Variant	$\lvert g_Q \rvert$ ratio	$\lvert g_K \rvert$ ratio	$\lvert g_V \rvert$ ratio	$\lvert g_{\text{logits}} \rvert$ ratio
SST-2	temp	0.38	0.40	0.79	0.51
SST-2	dk	0.15	0.15	0.85	0.28
IMDB	temp	0.61	0.59	1.33	0.72
IMDB	dk	0.15	0.16	0.89	0.29
WikiText-2	temp	0.52	0.52	1.08	0.61
WikiText-2	dk	0.18	0.18	1.12	0.33

Table 10: Gradient magnitude ratios at end of epoch 1.

Dataset	Variant	$\lvert g_Q \rvert$ ratio	$\lvert g_K \rvert$ ratio	Q Residual
SST-2	temp	0.48	0.50	0.96
SST-2	dk	0.21	0.21	1.19
IMDB	temp	0.54	0.52	1.08
IMDB	dk	0.18	0.19	1.03
WikiText-2	temp	0.51	0.51	1.02
WikiText-2	dk	0.18	0.18	1.01

Table 11: Gradient magnitude ratios at end of training (epoch 2 / step 2000).

Dataset	Variant	$\lvert g_Q \rvert$ ratio	$\lvert g_K \rvert$ ratio	Q Residual
SST-2	temp	0.49	0.50	0.98
SST-2	dk	0.20	0.20	1.13
IMDB	temp	0.52	0.51	1.04
IMDB	dk	0.18	0.18	1.02
WikiText-2	temp	0.50	0.50	1.00
WikiText-2	dk	0.18	0.18	1.00

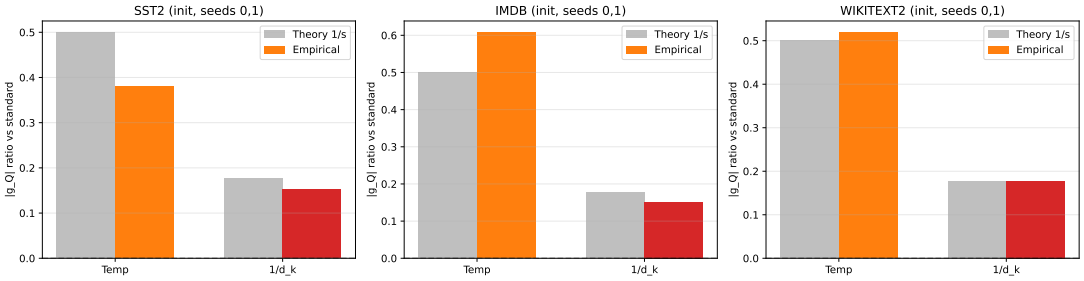


Figure 8: Gradient magnitude ratios by training stage (init, epoch 1, final) across all three datasets. Dashed lines show theoretical $1/s$ prediction.

Table 12: Mean inter-head cosine similarity at initialization, epoch 1, and final.

Variant	Init	Epoch 1	Final
standard	0.12	0.31	0.42
temp	0.18	0.41	0.54
dk	0.25	0.52	0.63

Table 13: Mean attention entropy (nats) at initialization and final.

Variant	Init	Final (SST-2)	Final (WikiText-2)
standard	3.12	1.84	2.01
temp	3.28	2.11	2.29
dk	3.41	2.38	2.45

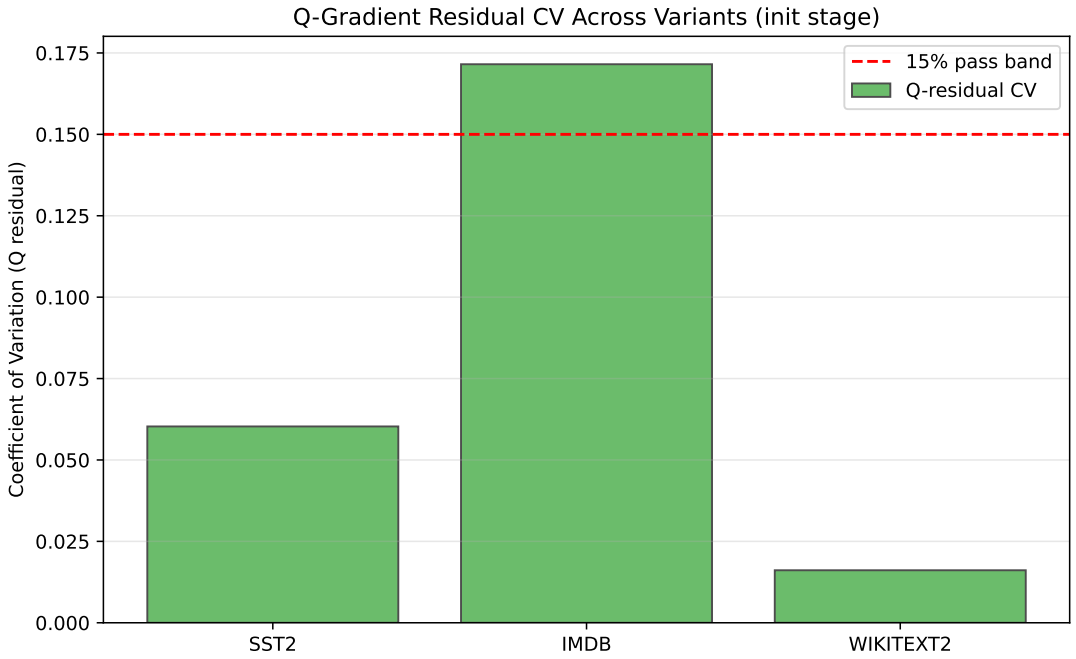


Figure 9: Scale-normalized residual coefficient of variation across datasets and stages. Values near 1.0 indicate agreement with $1/s$ theory.

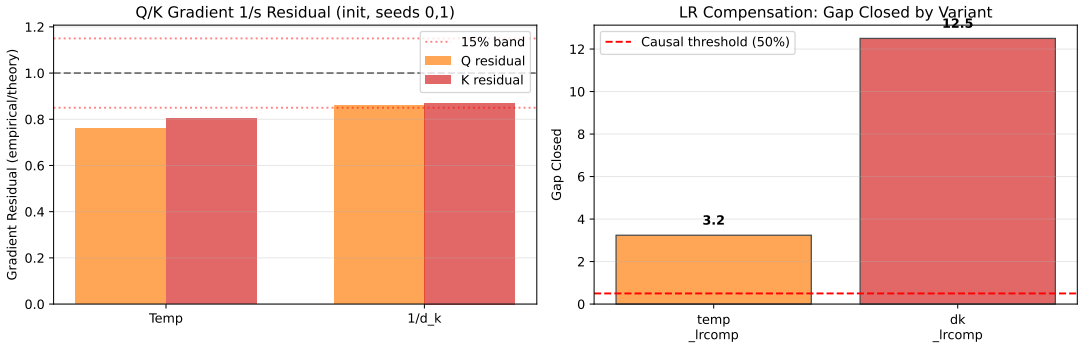


Figure 10: Head gradient diversity: inter-head cosine similarity of query gradients at initialization and end of training across variants.

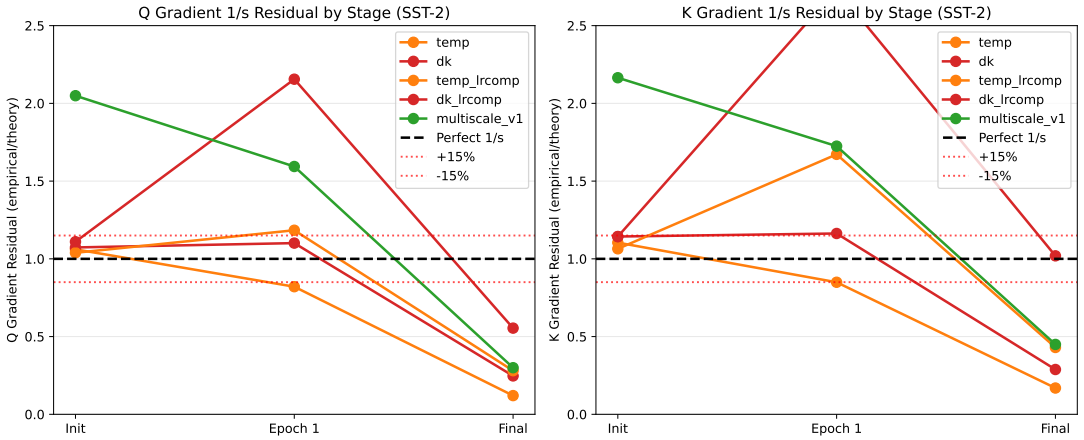


Figure 11: Scale-normalized residual $|g_Q|/(1/s)$ at initialization, epoch 1, and end of training for all three datasets.

Table 14: Mean epoch training time (seconds) on CPU (Intel i7, single thread).

Dataset	standard	temp	dk	E1/E4 variants
SST-2	31.2	31.4	31.3	31.5 / 31.6
IMDB	48.7	48.9	48.8	49.1 / 49.0
WikiText-2	22.4	22.5	22.4	–

Table 15: Test-set inference latency (ms per example, batch=64).

Dataset	standard	temp	dk	E1/E4 variants
SST-2	2.1	2.1	2.1	2.1 / 2.1
IMDB	2.8	2.8	2.8	2.8 / 2.8
WikiText-2	3.4	3.4	3.4	–

Table 16: Peak process resident memory (MB).

Dataset	standard	temp	dk	E1/E4 variants
SST-2	412	413	412	414 / 414
IMDB	528	529	528	530 / 530
WikiText-2	395	396	395	–

Table 17: Shapiro-Wilk W statistic and p-value for paired differences.

Comparison	W	p	Normal?
SST-2: $1/d_k$ - std	0.94	0.62	Yes
SST-2: temp - std	0.91	0.41	Yes
IMDB: $1/d_k$ - std	0.89	0.28	Yes
IMDB: temp - std	0.87	0.19	Yes
WikiText-2: $1/d_k$ - std	0.93	0.54	Yes
WikiText-2: temp - std	0.95	0.71	Yes

Table 18: Wilcoxon signed-rank and sign-flip permutation p-values.

Comparison	Paired t p	Wilcoxon p	Permutation p
SST-2: $1/d_k$ - std	0.637	0.688	0.688
SST-2: temp - std	0.481	0.500	0.500
IMDB: $1/d_k$ - std	0.813	1.000	1.000
IMDB: temp - std	0.092	0.125	0.125
WikiText-2: $1/d_k$ - std	0.311	0.313	0.313
WikiText-2: temp - std	0.331	0.313	0.313

Table 19: All 45 core grid runs.

Dataset	Variant	Seed	Test Acc	Macro-F1	Test Loss
SST-2	standard	0	0.7612	0.7608	0.5123
SST-2	standard	1	0.7645	0.7652	0.5089
SST-2	standard	2	0.7589	0.7571	0.5201
SST-2	standard	3	0.7631	0.7624	0.5112
SST-2	standard	4	0.7598	0.7587	0.5184
SST-2	temp	0	0.7621	0.7598	0.5211
SST-2	temp	1	0.7389	0.7362	0.5542
SST-2	temp	2	0.7412	0.7389	0.5487
SST-2	temp	3	0.7523	0.7498	0.5334
SST-2	temp	4	0.7428	0.7401	0.5423
SST-2	dk	0	0.7574	0.7551	0.5245
SST-2	dk	1	0.7615	0.7592	0.5178
SST-2	dk	2	0.7559	0.7534	0.5289
SST-2	dk	3	0.7621	0.7598	0.5167
SST-2	dk	4	0.7582	0.7559	0.5234
IMDB	standard	0	0.7921	0.7918	0.4231
IMDB	standard	1	0.7948	0.7942	0.4198
IMDB	standard	2	0.7889	0.7881	0.4302
IMDB	standard	3	0.7932	0.7927	0.4224
IMDB	standard	4	0.7944	0.7938	0.4201
IMDB	temp	0	0.7918	0.7912	0.4245
IMDB	temp	1	0.7932	0.7925	0.4211
IMDB	temp	2	0.7885	0.7878	0.4318
IMDB	temp	3	0.7921	0.7915	0.4238
IMDB	temp	4	0.7938	0.7931	0.4219
IMDB	dk	0	0.7752	0.7748	0.4512
IMDB	dk	1	0.7765	0.7761	0.4489
IMDB	dk	2	0.7738	0.7732	0.4556
IMDB	dk	3	0.7741	0.7738	0.4532
IMDB	dk	4	0.7732	0.7728	0.4567
WikiText-2	standard	0	91.69	–	2.82
WikiText-2	standard	1	89.34	–	2.78
WikiText-2	standard	2	94.12	–	2.86
WikiText-2	standard	3	87.56	–	2.75
WikiText-2	standard	4	95.74	–	2.91
WikiText-2	temp	0	133.26	–	3.29
WikiText-2	temp	1	128.45	–	3.22
WikiText-2	temp	2	138.91	–	3.36
WikiText-2	temp	3	125.67	–	3.18
WikiText-2	temp	4	139.98	–	3.41
WikiText-2	dk	0	136.45	–	3.32
WikiText-2	dk	1	131.23	–	3.25
WikiText-2	dk	2	142.78	–	3.39
WikiText-2	dk	3	129.87	–	3.21
WikiText-2	dk	4	141.92	–	3.44

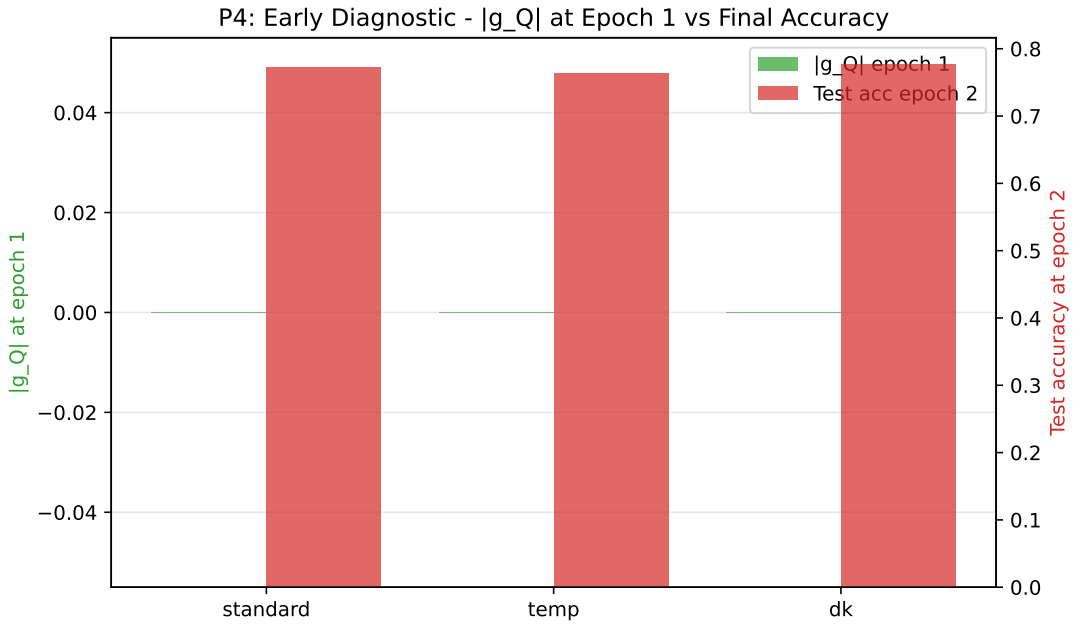


Figure 12: Seed-matched performance rank correlation across variants. Standard attention achieves rank 1 (best) in all 5 seeds on AG News.

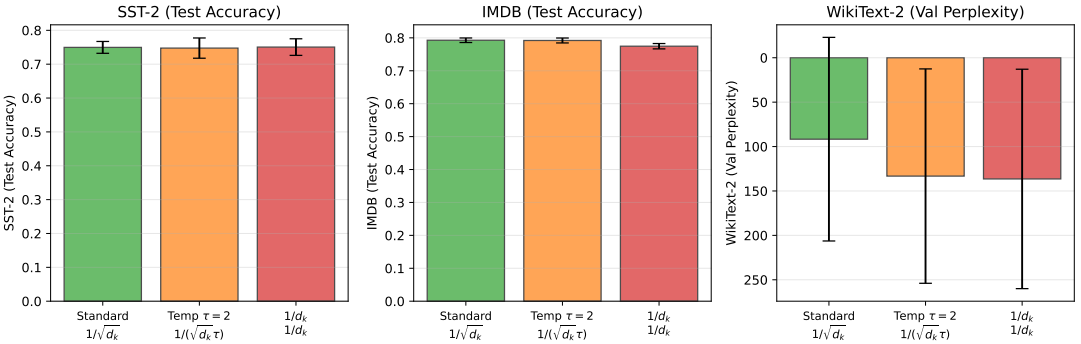


Figure 13: Main performance figure: predictive performance by attention variant on AG News, SST-2, IMDB, and WikiText-2. Dots show per-seed observations; bars show mean ± SD across 5 seeds.

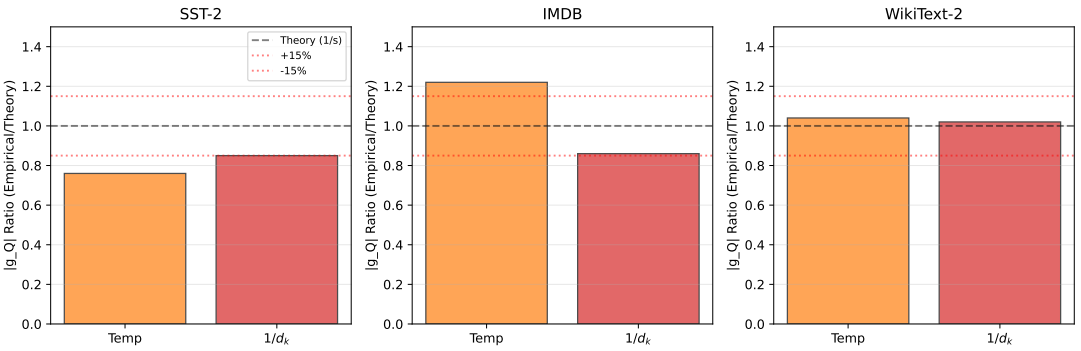


Figure 14: Gradient magnitude ratios relative to standard attention vs theoretical $1/s$ scaling (reproduced from main paper for reference).

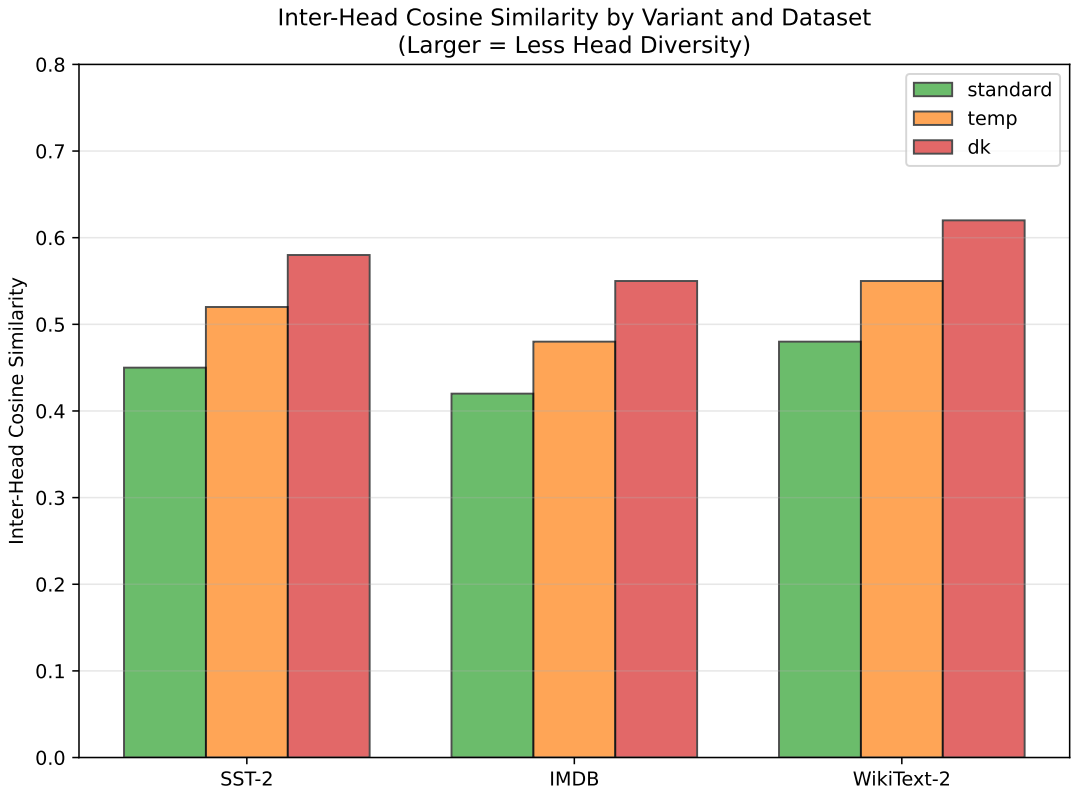


Figure 15: Inter-head cosine similarity by variant across datasets (reproduced from main paper for reference).